The school bus route

Optimization · Level 3 · Lesson 3



Today we will

- See why trying every bus route is impossible.
- Build a route with nearest neighbour and improve it with 2-opt.
- Judge a route by its length and by the longest ride.

Remember

- Routes multiply: 9 villages have 181,440, and 20 more than a billion billion.
- Nearest neighbour builds a route fast; 2-opt untangles it, but can stop short of the best.
- The children picked up first ride longest: agree a limit with the families.

The human bus route

1. Warm-up (5 min): Guess the shortest route by looking.
2. Nearest neighbour (10 min): The bus drives; count steps.
3. Untangle (10 min): Swap two crossing paths; count again.
4. Longest ride (10 min): Count each village’s ride.
5. Wrap-up (5 min): Shortest or fairest?

Spread the villages unevenly, with one far corner, so nearest neighbour leaves a long walk home. No playground? Use desks and a floor plan drawn on the board.

Let’s discuss

1. Why was the walk home so long with nearest neighbour?
2. Did swapping two crossing paths always help?
3. Which village rode longest, and is that fair?

Exit ticket

- Draw a route with a crossing, and the shorter route after one 2-opt swap.